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dGS: Direct Parameterization for Efficient Conditional Gaussian Splatting

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Verify Author List:  I have double-checked the author list and understand that additions and removals will not be allowed after the abstract submission deadline.

TL;DR: We introduce dGS, a direct parameterization for efficient conditional Gaussian splatting.

Abstract:

High-dimensional Gaussian Splatting (N-DGS) methods like 6DGS and 7DGS achieve impressive view- and temporal-dependent modeling through conditional N-dimensional Gaussians. However, their monolithic covariance design derives all conditioning effects from a joint covariance matrix, requiring expensive cubic-complexity matrix inversions in every forward pass. We introduce dGS, which replaces this implicit derivation with direct parameterization: opacity modulation is explicitly represented via Cholesky precision factors, and position shifts via displacement matrices. This eliminates matrix inversions while maintaining numerical stability. Our key contribution is learnable per-dimension coupling parameters that control how strongly position shifts respond to query deviations, enabling adaptive interpolation from query-independent positions to full N-DGS-style coupling. Across diverse benchmarks, dGS achieves up to 2.66x faster rendering on in-the-wild scenes (Mip-NeRF 360) while matching or exceeding N-DGS quality, offering a principled, efficient alternative to covariance-derived conditioning in high-dimensional Gaussian splatting.

Primary Area: Applications->Computer Vision

Keywords: 3D Gaussian Splatting, view dependent, novel view synthesis

Ethics Agreement:  I certify that all co-authors of this work have read and are committed to adhering to the Call for Papers, Author Instructions, Research Ethics, and Peer-review Ethics.

LLM Policy:  This submission allows Policy B.

Proceedings-only Option:  If this paper is accepted, the authors tentatively plan to present it in person at the conference (as a poster and, if selected, as an oral).

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Official Review of Submission1198 by Reviewer kTKd

Official Review by Reviewer kTKd 13 Mar 2026, 23:29 (modified: 24 Mar 2026, 08:21)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer kTKd

Revisions (/revisions?id=sD0GJIa1HY)

Summary:

This paper proposes dGS, a direct parameterization for conditional Gaussian splatting that decouples opacity and position conditioning from a monolithic N-dimensional covariance. Instead of deriving conditioning, dGS parameterizes a Cholesky factor of the precision for opacity and a bounded displacement matrix for position, with learnable per-dimension coupling Λ controlling how strongly position responds to query deviations. A CUDA implementation shows substantial rendering speedups while matching or surpassing the quality of 6DGS/7DGS across static and dynamic benchmarks.

Strengths And Weaknesses:

Strengths

1. Introduces a clean, efficient direct parameterization that removes the need for per-primitive matrix inversions in the conditioning step by using a Cholesky parameterization of precision for opacity and an explicit, spatially scaled displacement for position.

2. Evaluates on five datasets (NeRF Synthetic, Mip-NeRF 360, 6DGS-PBR, D-NeRF, 7DGS-PBR), with both dGS-O and full dGS variants.

Weaknesses

1. The paper relaxing the original probabilistic consistency of conditional Gaussian slicing. In N-DGS, conditional opacity, conditional mean, and conditional covariance are all derived from the same joint Gaussian. In dGS, opacity and position are directly parameterized, while covariance correction is removed entirely. As a result, the method no longer preserves the strict semantics of a single underlying conditional Gaussian model, and the paper does not fully convince me that is meaningful and correct.

2. The method does not clearly reduce end-to-end training cost. In Table 2, 3, full dGS is often similar to or slower than N-DGS in training time, so the claim 'Efficient' actually is not convincing. And also the claim of 'requiring expensive cubic-complexity matrix inversions in every forward pass' in N-DGS is somewhat overstated, since C is small (3 or 4), and triangular solves and caching can mitigate the cost.

3. Limited ablations on Learnable Coupling Control Λ : results do not disentangle learned Λ vs fixed $\Lambda=1$ vs $\Lambda=0$.

Soundness: 3: good

Presentation: 3: good

Significance: 2: fair

Originality: 2: fair

Key Questions For Authors:

please refer to the Strengths And Weaknesses

Limitations:

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2/7

A recent work, Universal Beta Splatting, has reported substantial gains in both reconstruction quality and training efficiency. Although it may be considered a parallel work and not strictly necessary for direct comparison, but its strong performance makes this paper less convincing.

Overall Recommendation: 3: Weak reject: A paper with clear merits, but also some weaknesses, which overall outweigh the merits. Papers in this category require revisions before they can be meaningfully built upon by others. Please use sparingly.

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

Compliance With LLM Reviewing Policy: Affirmed.

Code Of Conduct Acknowledgement: Affirmed.

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Official Review of Submission1198 by Reviewer hVDr

Official Review by Reviewer hVDr 13 Mar 2026, 01:02 (modified: 24 Mar 2026, 08:21)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hVDr
Revisions (/revisions?id=3YAeZovYh8)

Summary:
The method improves upon the 6DGS and 7DGS's gaussian query method by decomposing the big covariance matrix into an explicit displacement matrix for particle position, and a decay factor for opacity, accelerating rendering speeds.

Strengths And Weaknesses:
The idea is natural and intuitive. The mathematical derivation and algorithm design are smart while still appearing technically solid. The experimental results are also convincing.

Issues

1. In Figure 3, why are only a few regions zoomed in? Does dGS perform especially well in these selected regions?
2. The results are a bit difficult to parse visually, especially table 2 puts different methods vertically, while table 3 right below it puts different methods horizontally, and similar metrics are far away from each other.

Soundness: 3: good
Presentation: 3: good
Significance: 2: fair
Originality: 3: good

Key Questions For Authors:

1. There are other more complicated kernels now available, like beta splatting, spherical voronoi, textured gaussians etc. Would we have a comparison against these more expressive kernels to see where dGS stands?
2. The paper presents two variants: dGS-O and dGS, my impression is, dGS-O is just an ablation study, but it appears in all the main tables. What is the recommended method then? If there is a trade-off, when should which representation be used exactly?
3. It seems that dGS often achieves better quality. Why might this be the case? Intuitively, one might expect dGS to be less expressive than N-DGS.
4. dGS and dGS-O also seem to produce substantially more Gaussians in some settings. Is this a practical concern?

Limitations:
The paper has a Future Work section, but it does not really discuss limitations in a direct way. For example, does not explicitly discuss the increased Gaussian count in some settings, the practical trade-off between dGS-O and dGS, etc.

Overall Recommendation: 4: Weak accept: Technically solid paper that advances at least one sub-area of AI, with a contribution that others are likely to build on, but with some weaknesses that limit its impact (e.g., limited evaluation). Please use sparingly.

Confidence: 4: You are confident in your assessment, but not absolutely certain. It is unlikely, but not impossible, that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work.

Compliance With LLM Reviewing Policy: Affirmed.

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Official Review of Submission1198 by Reviewer ZkXw

Official Review by Reviewer ZkXw 📅 06 Mar 2026, 07:09 (modified: 24 Mar 2026, 08:21)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer ZkXw

📄 Revisions (/revisions?id=WKo6deEF40)

Summary:

This paper addresses the efficiency bottleneck in high-dimensional Gaussian Splatting methods (6DGS/7DGS, collectively N-DGS). N-DGS requires a matrix inversion per Gaussian primitive at every rendering pass to compute conditional opacity and position shifts, which becomes costly when primitives number in the millions. dGS proposes replacing this implicit derivation with three explicit parameterizations: (1) opacity decay via Cholesky precision factors, avoiding matrix inversion; (2) position displacement via displacement matrices scaled by Gaussian size; (3) learnable per-dimension coupling parameters $\Lambda \in [0,1]$ controlling how strongly positions respond to view/time queries. Experiments across five datasets show dGS achieves up to 2.66× speedup while matching or exceeding N-DGS quality.

Strengths And Weaknesses:

Strengths.

- The motivation is clear and addresses a real engineering problem. When a scene contains millions of Gaussian primitives, performing a matrix inversion for each one accumulates into non-negligible overhead. Replacing this step with lighter direct computation is a natural idea. Notably, the authors also implement equally optimized low-level kernels for the N-DGS baseline, ensuring a fair speed comparison — this is uncommon in the field and commendable.
- The learnable coupling parameter Λ is the most valuable design in this paper: it allows each Gaussian to independently decide whether its position should vary with view/time. When $\Lambda = 0$, positions remain fixed (opacity-only modulation); when $\Lambda = 1$, positions fully respond to query changes; intermediate values yield partial coupling. This is more flexible than N-DGS's rigid coupling structure. The analysis in Appendix A is also convincing, clearly showing how the implicit parameterization in N-DGS leads to asymmetric displacement bounds depending on spatial dimension ordering.
- The experiments cover five datasets (three static, two dynamic) with two densification strategies, providing reasonable diversity.

Weaknesses.

- The true source of the speedup is unclear. The paper's core narrative is that eliminating matrix inversion brings the speedup, but the matrices being inverted are tiny (3×3 or 4×4), and a single inversion is computationally negligible. The real reason dGS is faster likely lies primarily in the fact that its computation pattern is more amenable to efficient GPU execution (e.g., triangular matrix-vector products fuse more naturally into a single compute step than general matrix inversions), rather than a fundamental algorithmic complexity advantage. The paper repeatedly emphasizes " $O(C^3)$ complexity," but when $C = 3$ or 4 , C^3 amounts to merely 27 or 64 operations — the asymptotic analysis is substantially overstated in practice. A simple controlled experiment — swapping only the direct computation back to N-DGS-style matrix inversion within the same code framework, keeping everything else identical — would definitively clarify this.
- The comparison scope is too narrow. The paper only compares against N-DGS (6DGS/7DGS), a single method family. There are many alternative approaches for modeling view-dependent effects and dynamic scenes — neural network-based methods like Scaffold-GS and GaussianShader for view dependence, and deformation-

based methods like Deformable 3D Gaussians and SC-GS for dynamics. Comparing against only one method family is insufficient to establish competitiveness within the broader method space. If these alternatives already avoid matrix inversion and achieve better quality, the value of accelerating N-DGS becomes limited.

- The primitive count inflation problem. A careful reading of Tables 3 and 6 reveals that dGS/dGS-O uses far more Gaussian primitives than N-DGS on several datasets. The most extreme case is the stump scene in Mip-NeRF 360: N-DGS uses 644K primitives while dGS uses 3343K — over 5×. The Mip-NeRF 360 average is roughly 2× (2463K vs 1273K). This suggests that direct parameterization has less expressive power than the implicit approach and requires more primitives to compensate. Yet the paper not only fails to discuss this issue, but interprets "using more primitives while still being faster" as an advantage. More primitives mean greater memory overhead and longer training times, which are real costs in resource-constrained settings.

Soundness: 2: fair

Presentation: 3: good

Significance: 2: fair

Originality: 2: fair

Key Questions For Authors:

1. Speedup attribution: Within the same code framework, could you replace only dGS's direct computation with N-DGS-style matrix inversion (keeping everything else identical) and report the speed difference? This would directly answer how much of the speedup comes from eliminating matrix inversion itself.
2. Controlled primitive count comparison: If dGS's primitive count is capped at the same level as N-DGS (e.g., ~1273K for Mip-NeRF 360), how do quality and speed change? This would clarify whether primitive inflation is necessary for dGS to maintain quality.
3. Comparison with alternative approaches: Could you add comparisons with at least one method that does not rely on high-dimensional Gaussians (e.g., Scaffold-GS or Deformable 3D Gaussians)?

Limitations:

No. The paper lacks a Limitations section. I recommend adding discussion of: the applicability limits due to dropping the covariance correction term, the causes and costs of primitive count inflation, and the scope of the block-diagonal precision matrix assumption.

Overall Recommendation: 3: Weak reject: A paper with clear merits, but also some weaknesses, which overall outweigh the merits. Papers in this category require revisions before they can be meaningfully built upon by others. Please use sparingly.

Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

Compliance With LLM Reviewing Policy: Affirmed.

Code Of Conduct Acknowledgement: Affirmed.

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Official Review of Submission1198 by Reviewer BMak

Official Review by Reviewer BMak  04 Mar 2026, 09:21 (modified: 24 Mar 2026, 08:21)

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer BMak

 Revisions (/revisions?id=lcKmhRxJRt)

Summary:

The paper proposes dGS, a direct parameterization for conditional Gaussian splatting that replaces covariance-derived conditioning used in N-D Gaussian splatting (e.g., 6DGS and 7DGS). Instead of computing view- or time-dependent effects through conditional Gaussian formulas requiring matrix inversion, the method directly parameterizes opacity modulation and position displacement using Cholesky precision factors and displacement matrices. A learnable coupling parameter controls how strongly position shifts depend on query variables. Experiments on static and dynamic benchmarks show that the method achieves comparable or better reconstruction quality while providing up to 2.66× faster rendering.

Strengths And Weaknesses:

Soundness.

The proposed method is technically sound and well motivated. The paper clearly identifies the computational bottleneck in N-DGS approaches, namely the repeated inversion of covariance submatrices required for conditional Gaussian evaluation. The proposed direct parameterization replaces this expensive step with a more efficient formulation. The derivation is straightforward and preserves the behavior of conditional Gaussian conditioning while avoiding matrix inversions.

The experimental evaluation includes both static and dynamic datasets, however baseline selection is very limited. Results seem to be strong but more comparisons are needed to fully evaluate them. The ablation studies validate the importance of key components such as spatial scaling and position displacement.

Presentation.

The paper is clearly written and well structured. Motivation is clear. The appendix provides additional clarification on the relationship between the proposed displacement formulation and the implicit covariance-based parameterization used in N-DGS. The inclusion of implementation details, CUDA optimization discussion, and extensive experiments further supports reproducibility.

Importance.

The work addresses an important efficiency challenge in Gaussian splatting-based radiance field representations. As these methods should scale to larger scenes and more complex conditioning variables such as view direction and time, improving computational efficiency becomes increasingly important.

Originality. I think this paper could meet the bar for acceptance if the raised questions are addressed (see box below). The proposed method introduces a direct parameterization perspective for conditional Gaussian splatting by decoupling conditioning effects from the covariance matrix. While the individual components are known, their combination into a unified framework for conditional Gaussian slicing represents a meaningful conceptual contribution if the efficiency and results can be supported with additional experiments.

Soundness: 3: good

Presentation: 3: good

Significance: 3: good

Originality: 3: good

Key Questions For Authors:

1. **Limited real-world dynamic evaluation.** Although the experiments cover several datasets, evaluation on challenging **real-world** dynamic scenes appears limited. Testing on more in-the-wild dynamic datasets could further demonstrate the robustness of the approach. For example: NeRF-DS[1]. I suggest this dataset since its 1) dynamic 2) includes specular (view-dependent) effects 3) real world. Interesting insights could be shown here. I guess similar experiments could be conducted on [2] on scenes like "Flame Salmon". These are just examples, not exhaustive list. Also, for dynamic scenes authors should include videos and evaluate if potential artifacts appear there (you can clearly see this from images).
2. **Very limited baseline selection.** The idea of higher dimensional Gaussians is not very recent, and authors only compare their work with N-DGS. Examples [3; 4-also mentioned in related works in the paper]. These are examples, not the exhaustive list.

Non-technical remark: why full results for other baselines (4DGS, 3DGS) are not included in Table 3? These are valid baselines and should be in the table 3.

For comparisons with 3DGS (Kerbl et al., 2023) and 4DGS (Yang et al., 2024) baselines, refer to the original N-DGS (Gao et al., 2025a;b).

It is very inconvenient for the reader to look in another paper for comparison. There is space, or authors could include the results in supplementary material if definitely could not fit the main paper.

2. **Failure cases are missing.** are there any failure cases typical for this method?
3. **Scalability analysis.** The paper focuses on efficiency improvements but provides limited analysis of scalability when additional conditioning variables are introduced.

4. What is the **memory overhead** of the additional explicit parameters compared to the covariance representation used in N-DGS?

[1] NeRF-DS: Neural Radiance Fields for Dynamic Specular Objects by Zhiwen Yan, Chen Li, Gim Hee Lee, CVPR 2023.
[2] Neural 3D Video Synthesis from Multi-view Video by Tianye Li1,3,* Mira Slavcheva1,* Michael Zollhoefer1 , CVPR 2022.
[3]4D-Rotor Gaussian Splatting: Towards Efficient Novel View Synthesis for Dynamic Scenes by Yuanxing Duan1
Fangyin Wei3 Qiyu Dai1,2, SIGGRAPH2024.
[4]6DGS: Enhanced Direction-Aware Gaussian Splatting for Volumetric Rendering

Limitations:
The paper does not include a dedicated Limitations section and instead only discusses future work. Some potential limitations could be inferred from my questions raised above; depending on the authors’ responses, these points could be incorporated as an explicit discussion of limitations in the paper.

For now I am assigning the paper **weak reject** because of the listed questions. But I am open to raise the score in the discussion period if authors address my questions.

Overall Recommendation: 3: Weak reject: A paper with clear merits, but also some weaknesses, which overall outweigh the merits. Papers in this category require revisions before they can be meaningfully built upon by others. Please use sparingly.
Confidence: 3: You are fairly confident in your assessment. It is possible that you did not understand some parts of the submission or that you are unfamiliar with some pieces of related work. Math/other details were not carefully checked.

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